Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN Practical

Practical – II

Roll No. : T048	Name: Lavkush Vishwakarma
Class : TYIT	Date/Time of Submission : 30/06/2026

Aim: Core JavaScript Concepts for MERN.

1.A). Create functions and call them with parameters.

SOURCE CODE:-

```
function calculateSalary(salary, name) {
    let bonus = 5000;
    let totalSalary = salary + bonus;
    console.log("Employee Name: " + name);
    console.log("Base Salary: " + salary);
    console.log("Bonus: " + bonus);
    console.log("Total Salary: " + totalSalary);
}
calculateSalary(30000, "John Doe");
2nd
function calculate(product, quantity, price) {
    let total = quantity * price;
    let gst = total * 18 / 100;
    let finalAmount = total + gst;

    console.log("Product:", product);
    console.log("Quantity:", quantity);
    console.log("Price:", price);
    console.log("Total:", total);
    console.log("GST:", gst);
    console.log("Final Amount:", finalAmount);
}
calculate("Book", 5, 100);
console.log("Lavkush T048");
```

OUTPUT:-

```
PS D:\Users\Lavkush\Study\MERN> node P2.js
Employee Name: John Doe
Base Salary: 30000
Bonus: 5000
PS D:\Users\Lavkush\Study\MERN> node P3.js
Product: Book
Quantity: 5
Price: 100
Total: 500
GST: 90
Final Amount: 590
Lavkush T048
```

1.B). Work with arrays and objects in JavaScript.

SOURCE CODE:-

```
let fruits = ["Apple", "Banana", "Mango", "Orange"];
let student = {
    name: "John Doe",
    age: 20,
    class: "12th Grade",
};
console.log("Fruits:", fruits);
console.log("Second Fruit:", fruits[1]);
console.log("Third Fruit:", fruits[2]);
console.log("Student Name:", student.name);
```

```
console.log("Student Age:", student.age);
console.log("Student Class:", student.class);
console.log("Lavkush T048");
```

OUTPUT:-

```
PS D:\Users\Lavkush\Study\MERN> node P4.js
Fruits: [ 'Apple', 'Banana', 'Mango', 'Orange' ]
Second Fruit: Banana
Third Fruit: Mango
Student Name: John Doe
Student Age: 20
Student Class: 12th Grade
Lavkush T048
```

1.C). Demonstrate use of arrow functions.

SOURCE CODE:-

```
let add = (a, b) => {
   return a + b;
};
let sub = (a, b) => {
   return a - b;
};
let mul = (a, b) => {
   return a * b;
};
let div = (a, b) => {
   return a / b;
};
console.log("Addition: ", add(34, 96));
console.log("Subtraction: ", sub(34, 96));
console.log("Multiplication: ", mul(34, 96));
console.log("Division: ", div(34, 96));
console.log("Lavkush T048");
```

OUTPUT:-

```
PS D:\Users\Lavkush\Study\MERN> node P5.js
Addition:  130
Subtraction:  -62
Multiplication:  3264
Division:  0.3541666666666667
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```

1.D). Write a program using array methods (map, filter, reduce).

SOURCE CODE:-

```
let numbers = [10,20,30,40,50]
let doubled = numbers.map(num => num * 2);
let greater25 = numbers.filter(num => num > 25);
let total = numbers.reduce((sum, num) => sum + num, 0);
console.log("Orignal Array:",numbers);
console.log("Map:",doubled);
console.log("Filter:",greater25);
console.log("Reduce:",total);
console.log("Lavkush T048");
```

OUTPUT:-

```
PS D:\Users\Lavkush\Study\MERN> node P6.js
Orignal Array: [ 10, 20, 30, 40, 50 ]
Map: [ 20, 40, 60, 80, 100 ]
Filter: [ 30, 40, 50 ]
Reduce: 150
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```

1.E). Create a program that manipulates objects and displays output.

SOURCE CODE:-

```
let student = {
   name: "Zeus",
   rollno: 14,
   age: 19,
   marks: 95
};
console.log("Orignal Object:");
console.log(student);
//Accessing Properties
console.log("Accessing Properties")
console.log("Name:",student.name);
console.log("Marks:",student.marks);
//Updating  A Property
student.marks = 99;
console.log("After updating")
console.log(student);
//Adding A New Property
student.city = "Mumbai";
console.log("After Adding A New Property ")
console.log(student);
//Delete A Property
delete student.rollno;
```

```
console.log("After Deleting A Property ")
console.log(student);
console.log("Lavkush T048");
```

OUTPUT:-

```
PS D:\Users\Lavkush\Study\MERN> node P7.js
Orignal Object:
{ name: 'Zeus', rollno: 14, age: 19, marks: 95 }
Accessing Properties
Name: Zeus
Marks: 95
After updating
{ name: 'Zeus', rollno: 14, age: 19, marks: 99 }
After Adding A New Property
{ name: 'Zeus', rollno: 14, age: 19, marks: 99, city: 'Mumbai' }
After Deleting A Property
{ name: 'Zeus', age: 19, marks: 99, city: 'Mumbai' }
Lavkush T048
```

1.F). Write programs using loops and template literals.

SOURCE CODE:-

```
let students = [
   {
      name: "Zeus",
      rollno: 14,
      age: 19,
      marks: 95
   },
   {
      name: "Bhavesh",
      rollno: 39,
      age: 19,
      marks: 91
   },
   {
      name: "Lavkush",
      rollno: 48,
      age: 19,
      marks: 94
   }
];
function display(s) {
   console.log("Student Details:")
   for (let stud of s) {
      console.log(`Name : ${stud.name}`);
      console.log(`Age : ${stud.age}`);
      console.log(`Roll no : ${stud.rollno}`);
      console.log(`Marks : ${stud.marks}`);
   }
}
display(students)
console.log("Lavkush T048");
```

OUTPUT:-

```
PS D:\Users\Lavkush\Study\MERN> node P8.js
Student Details:
Name : Zeus
Age : 19
Roll no : 14
Marks : 95
Name : Bhavesh
Age : 19
Roll no : 39
Marks : 91
Name : Lavkush
Age : 19
Roll no : 48
Marks : 94
Lavkush T048
```